



Static hand gesture recognition method based on the Vision Transformer

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Abstract

Hand gesture recognition (HGR) is the most important part of human-computer interaction (HCI). Static hand gesture recognition is equivalent to the classification of hand gesture images. At present, the classification of hand gesture images mainly uses the Convolutional Neural Network (CNN) method. The Vision Transformer architecture (ViT) proposes not to use the convolutional layers at all but to use the multi-head attention mechanism to learn global information. Therefore, this paper proposes a static hand gesture recognition method based on the Vision Transformer. This paper uses a self-made dataset and two publicly available American Sign Language (ASL) datasets to train and evaluate the ViT architecture. Using the depth information provided by the Microsoft Kinect camera to capture the hand gesture images and filter the background, then use the eight-connected discrimination algorithm and the distance transformation algorithm to remove the redundant arm information. The resulting images constitute a self-made dataset. At the same time, this paper studies the impact of several data augmentation strategies on recognition performance. This paper uses accuracy, F1 score, recall, and precision as evaluation metrics. Finally, the validation accuracy of the proposed model on the three datasets achieves 99.44%, 99.37%, and 96.53%, respectively, and the results obtained are better than those obtained by some CNN structures.

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1 Introduction

With the rapid development of artificial intelligence and computer vision, human-computer interaction has increased extensively in recent decades. Hand gesture information is more accessible than sounds, brain electrical signals, and other information, so hand gesture recognition has become one of the principal directions of human-computer interaction. Hand gesture recognition usually is divided into static hand gesture recognition and dynamic hand gesture recognition. There are two main methods for static hand gesture recognition. The first is hand gesture recognition based on wearable devices, such as data gloves [23] and electromyography [35]. However, it has considerable inconvenience and high cost.

In contrast, hand gesture recognition based on computer vision is more convenient. It only needs to use the camera to obtain hand gesture images without wearing additional equipment. The vision-based hand gesture recognition methods can divide into hand-crafted and deep learning methods. Hand-crafted methods include complex visual processing, such as hand-crafted feature extraction methods and classification algorithms. However, deep learning methods use supervised learning to adapt to more diverse and complex challenges. Convolutional Neural Network (CNN) is one of the deep learning methods and the most important one. As a result, most of the current papers involving hand gesture recognition use CNN-based model structures [2, 33, 37, 38, 42], such as AlexNet [20], VGG-16, VGG-19 [40], GoogLeNet [41], ResNet-50 [15].

At the same time, a new type of deep learning model used in natural language processing (NLP) was proposed, which is called the Transformer. Furthermore, it has appeared in some popularity in computer vision, such as image classification [14] tasks. Initially, it was introduced to solve machine translation [49], language modelling [11] and speech recognition [8]. And it successfully replaced the use of recurrent models (such as RNNs, LSTMs) in processing sequence data. The key to the Transformer's success is to rely on a mechanism called self-attention, which can capture the internal correlation of data and effectively solve the problem of long-distance dependence. In visual tasks, images are often processed; however, the number of pixels in the image is much larger than the number of words in the sequence task, so the attention mechanism is limited in the visual task. Hu et al. [16] proposed channel attention and SENet for this purpose. The core of SENet is a squeeze-and-excitation (SE) block to collect global information, capture channel relationships, and improve representation. To increase the receptive field, Wang et al. [50] proposed a non-local block and introduced self-attention to computer vision. Huang et al. [17] proposed CCNet, in which the self-attention operation is regarded as graph convolution, and the densely-connected graph processed by self-attention is replaced by several sparsely-connected graphs. To this end, it proposes criss-cross attention, which considers row attention and column attention recurrently to obtain global information. Recently, a model called the Vision Transformer (ViT) [14] was proposed by Dosovitskiy et al. It divides the image into multiple patches to be input to the encoder module of the attention model, and there are minor changes to the Transformer model, and this is a groundbreaking method. It is trained on a large amount of data and computing resources, and the ViT architecture has been proved to be more accurate in image classification tasks than the CNN architecture with the same parameters [14].

Much research shows that under the premise of using enough data for training, the performance of the ViT architecture is better than that of the CNN architecture in many aspects. Firstly, under the premise of pre-trained on the ImageNet dataset and having the same error rate, the ViT architecture has better transferability to downstream tasks than the CNN architecture [53]. Secondly, the ViT architecture has strong robustness under severe occlusions, disturbances, and domain shifts compared with the CNN architecture [6, 28, 32].

This paper aims to train a classifier that can accurately recognize static hand gesture images to complete the recognition of static hand gestures. Because the ViT structure has a better effect on image classification than the CNN structure, the transferability and robustness of the ViT structure on downstream tasks are better than the CNN structure. Therefore, this paper recognizes static hand gestures based on the pre-trained ViT structure. This paper applies the data augmentation strategies of the CutOut and the CutMix to improve the recognition accuracy. In the end, under the three hand gesture datasets, the results obtained by applying the ViT structure of this paper are better than most of the results obtained by applying the CNN structures.

The remainder of this paper is structured as follows. In Section 2, some recent related works on gesture recognition are summarized. The methods proposed in this paper mainly include foreground segmentation, arm removal, the Vision Transformer architecture, and data augmentation strategies, which are all described in Section 3. Section 4 introduces the entire experiment on this paper and the experimental results on three datasets. Then finally, Section 5 shows the conclusion of this paper.

2 Related works

The research methods of hand gesture recognition are roughly divided into methods based on data gloves and electromyography, methods based on hand-crafted feature extraction, methods based on computer vision, and combinations based on the three methods. The following describes the recent related work on hand gesture recognition.

In the methods based on hand-crafted feature extraction, Pan et al. [31] proposed to use Gaussian Mixture Model (GMM) to segment the hand region of RGB images. Then the segmented images were processed by Fourier Descriptor, Hu moments and Scale-invariant feature transform (SIFT) to extract independent features. Before being classified by Support Vector Machine (SVM), PCA and LDA were used to trim the dimensionality of the feature vector. Among the many dynamic hand gesture recognition methods, Bendarkar et al. [3] proposed a real-time gesture recognition system based on a combination of static and dynamic hand gestures based on convolutional neural networks (CNN). In the dynamic hand gesture recognition part, a bidirectional ConvLSTM and 3DCNN were combined to form a 3DCNN-LSTM model.

Oyedotun and Khashman [29] proposed three different CNNs and SDAEs architectures and applied them to Thomas Moeslund's gesture recognition dataset for evaluation. After the final training, the accuracy of CNN1 and SDAE3 on the test data achieved 91.33% and 92.83%, respectively.

Chevtchenko et al. [10] proposed a real-time hand gesture recognition architecture based on feature fusion convolutional neural network (FFCNN), which combined Convolutional Neural Network (CNN) and Gabor filters for feature extraction and gesture classification, and the Tree-of-Parzen-Estimators (TPE) algorithm was used to search the hyperparameters of the network, which was introduced by Bergstra et al. [4]. It was evaluated on three different benchmark datasets using binary, gray, and depth data and two different

validation techniques. Moreover, the system recognized gestures from external operators and LaRED datasets in real time. Ozcan and Basturk [30] proposed a gesture recognition architecture based on the AlexNet model in the later work. The hyperparameters of the architecture were optimized by a heuristic optimization algorithm based on artificial bee colonies.

The Microsoft Kinect camera can provide three types of data: skeleton, depth, and color, so the camera has been widely used to collect gesture data. Li and Tang et al. [21] proposed to use the Microsoft Kinect camera to acquire depth and color images, and a joint bilateral filtering algorithm was used to denoise the depth images. Then the depth information was used to segment the hand area in the color image. A convolutional neural network was used to complete the feature extraction part, and finally, a support vector machine (SVM) was applied as a classifier. In the end, the network achieved 98.52% accuracy on eight gestures.

Tao et al. [45] proposed a CNN to realize American Sign Language letter recognition and used multi-view image augmentation and inference fusion. The multi-view augmentation strategy referred to using the Microsoft Kinect camera to capture a gesture depth image and 3D point cloud. Then set additional virtual cameras to face the point cloud of different perspectives so that a set of additional images were generated, and these images were subjected to some preprocessing, and the unnecessary arm information was removed, and finally, a depth image for training was obtained. In the inference step, the method included information from multiple views for the final prediction to solve the confusion caused by the orientation change and partial occlusion. On the two benchmark datasets, the most advanced methods were achieved.

Nagi et al. [27] proposed a deep neural network that combines convolution and max-pooling operations (MPCNN). Gesture images were collected by humans wearing colored gloves and using mobile robots and then extracted to the hand area using color segmentation, performed morphological operations, and eliminated noise. The network classification accuracy rate of the six gestures achieved 96%. And the robot using an ARM 11 533MHz processor, realized real-time hand gesture recognition.

Mohammed et al. [26] proposed a deep learning-based architecture to jointly detect gestures and classify gestures. The architecture used a one-stage dense object detector RetinaNet to extract the hand area and a lightweight convolutional neural network MobileNet for gesture recognition. The method of the hand gesture detection part was tested on four public datasets, including the Oxford, 5-signers, EgoHands, and Indian classical dance (ICD) datasets, and the average precisions (AP) obtained were 72.10% and 97.90%, 93.10%, and 85.50%, respectively. The method of hand gesture recognition part was evaluated on the LaRED and the simple and complex TinyHands datasets, and the accuracy rates obtained were 97.25%, 99.00%, and 90.43%.

All in all, the vast majority of the above works are based on the CNN architecture, and the datasets used are relatively single. The ViT-based model proposed in this paper does not require the CNN architecture at all. Simultaneously, multiple datasets including publicly available self-made dataset are used to verify the availability of the model. For the self-made dataset in this paper, the depth information of the Kinect V2 camera is used to complete the foreground segmentation of the gesture image, and then the forearm region in the gesture image is removed, thereby reducing unnecessary features in the process of gesture feature extraction. At the same time, the above related works all use standard data augmentation strategies to varying degrees. This paper explores the impact of different data augmentation strategies on gesture image classification.

3 Proposed method

3.1 Foreground segmentation

As we know, the Kinect V2 camera released by Microsoft provides not only RGB images of traditional cameras but also depth and human skeleton information, and the camera can simultaneously capture these three kinds of information [44]. The depth information retains the three-dimensional shape of the object and has invariance to the difference in lighting and the change of the background. The Kinect V2 allows remote interaction between the hand and the camera within a range of 0.5 m to 4.5 m.

At the same time, the Kinect V2 can map the depth information to the color image. According to this, the camera can separate the hand region of interest from the background. The depth threshold is used to filter out the background. Through the corresponding relationship between the depth information and the color information, a color image of the foreground containing only the region of interest of the hand is obtained. The hand region of interest is extracted as shown in Fig. 1. As shown in Fig. 1(b), after the above method, only the image of the difference Δz (the value of Δz is 350 mm) between the maximum threshold and the minimum threshold is retained, thereby eliminating the human body and the background part, and obtaining the gesture image A containing the arm part.

3.2 Arm removal

To reduce the calculation time when extracting features and classifying gestures, removing unwanted forearm regions from the image is necessary. The arm removal method in this paper is based on the distance transformation algorithm and eight-connected discrimination algorithm [51]. The distance in the distance transformation algorithm refers to Euclidean distance.

Figure 2 shows the complete flow of the arm removal algorithm. It consists of the following steps:

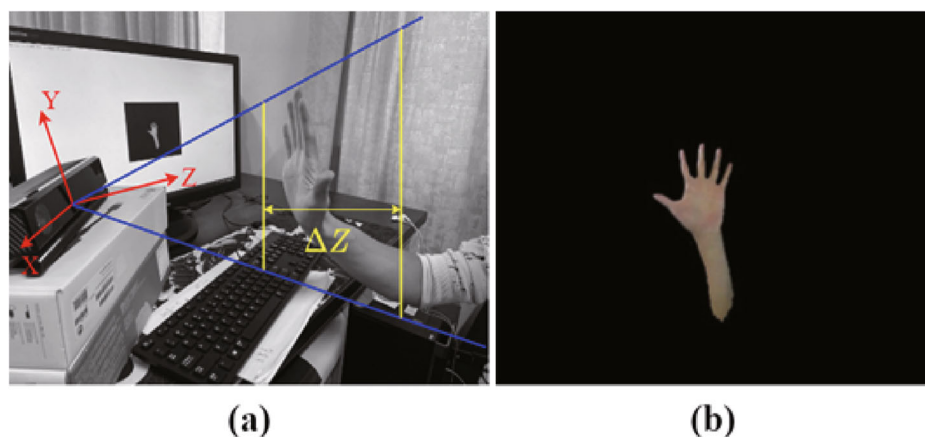


Fig. 1 Extracted hand region of interest

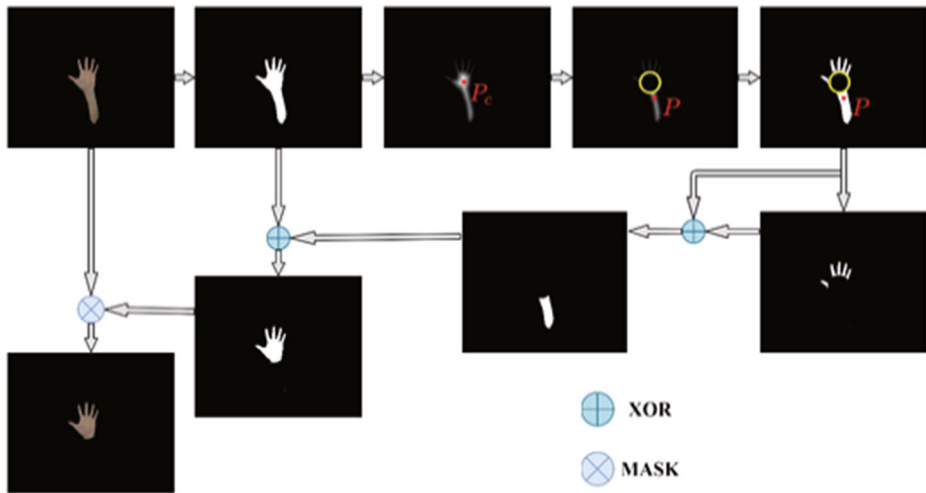


Fig. 2 The process of arm removal

Step 1: Use the gesture image A after foreground segmentation in the previous section, apply the Otsu algorithm to threshold segmentation to obtain the binary image H of the gesture.

Step 2: The distance transformation operation is performed on the obtained binary image H . Then, image D can be obtained. The pixel value in image D is represented as the distance from the pixel point inside the hand to the boundary of the hand, the maximum value is generally located in the center of the palm (pixel point P_c in Fig. 2), and the maximum value is regarded as the radius R_0 of the inscribed circle of the palm.

Step 3: Draw a circle on the obtained image H and image D . To keep the palm in the circle (yellow circle in the Fig. 2), select P_c as the center, $R_1 = 1.35 \times R_0$ is selected as the radius, and then make the pixel values in the circle all 0 to get an image with the palm part removed.

Then the maximum pixel value of the remaining area in the image D is detected and take it as P_{max} , which corresponds to the pixel point P . When there is an arm area in the image, P is usually located at the midline of the arm (pixel point P in the Fig. 2), and P_{max} is relatively large; when there is no arm area in the image, P is generally located at the midline of the finger, and P_{max} is relatively small. Because the gesture data collected in this study all include arms, the situation without arms is not considered.

Step 4: Use the eight-connected discrimination algorithm to find the label values of all areas. Suppose the label value of the currently connected area is the same as the label value of the pixel point P . The pixel value of this part is replaced with 0, and the resulting binary image contains only fingers and the palm of the part where the fingers are connected.

Step 5: Perform an XOR operation between the results obtained in Step 3 and Step 4, the binary image containing only the arm area is obtained.

Step 6: Perform an XOR operation between the result obtained in Step 5 and the binary image H of the original hand to obtain the binary image of the gesture after removing the arm.

Step 7: Perform a mask operation between the binary image obtained in Step 6 and the

original collected RGB image to obtain the final gesture RGB image that contains only the fingers and palm after the arm is removed.

Compared with other arm removal algorithms [24, 25, 39], this algorithm can remove the arm effectively and quickly, and is not affected by the rotating hand.

3.3 Vision Transformer

The Vision Transformer is a computer vision model based entirely on the Transformer architecture [48] originally designed for text tasks. The Vision Transformer was proposed to extend the standard Transformer to computer vision tasks. The Transformer architecture has shown the best performance in machine translation and other NLP tasks in recent years, so that it has attracted the interest of many people [12]. For image classification tasks, everyone thinks of the Convolutional Neural Networks (CNN), such as VGGNets [40]. The traditional CNN architectures usually use filters with local receptive fields. The Vision Transformer is a completely different architecture from the traditional CNN, and it has a more unified representation in all layers. In addition, the method of the Vision Transformer for image classification is to use the encoder module of the Transformer to map a series of image patches to the semantic labels. After exploration [36], it is found that the difference between the Vision Transformer and the CNN is that its self-attention mechanism plays a key role, and its mechanism can enable the early aggregation of global information. And the multi-head self-attention module in the Vision Transformer also contains residual connections, which can learn better global information.

The overall system architecture is shown in Fig. 3. The system architecture can be composed of a linear embedding layer, an encoder, and a final classifier. Before entering the linear embedding layer, the image X of size $c \times h \times w$ (where c represents the number of channels of the image, h is the height of the image, and w is the image's width) from the training set needs to be subdivided into non-overlapping patches. Then each patch is treated as a separate token in the Transformer. Each patch of size $c \times p \times p$ is extracted from the image, thus forming a series of patches $(x_1, x_2, \dots, x_n)$ of length n , where $n = hw/p^2$. Usually, the $p \times p$ is 16×16 or 32×32 , where a smaller patch resolution will lead to a longer sequence.

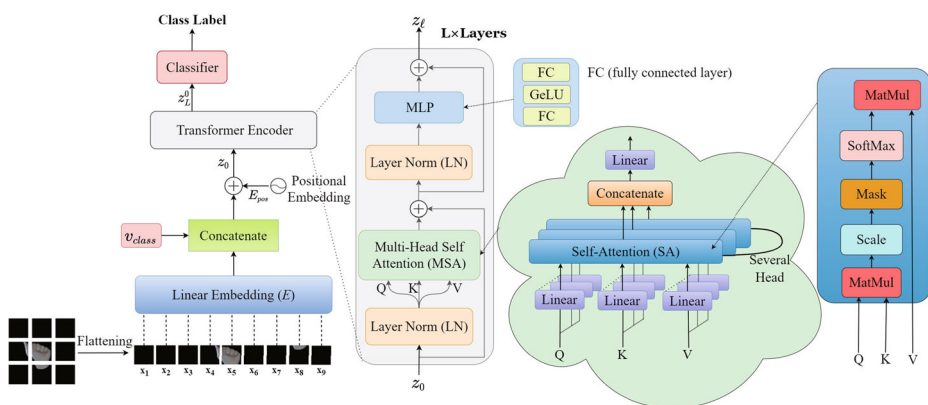


Fig. 3 The architecture of the Vision Transformer includes the main architecture of the model, the transformer encoder module, the multi-headed self-attention (MSA), and the self-attention (SA) module

3.3.1 Linear embedding layer

Because for the standard Transformer structure, the sequence of token must be entered. However, the data structure of the above patches is a three-dimensional matrix, so before the sequence of patches is sent to the encoder, the learnable embedding matrix E is used to project it into the vector of the model dimension d . This is the role of the linear embedding layer.

The output of the linear embedding layer needs to be concatenated with the learnable classification mark v_{class} required to perform the classification task. To keep the spatial arrangement of the patches the same as the original image, the encoded position information E_{pos} is added to the patch representations. Finally, an embedding sequence with token z_0 is generated. As shown in the Eq. (1):

$$z_0 = [v_{class}; x_1 E; x_2 E; \dots; x_n E] + E_{pos}, E \in \mathbb{R}^{(p^2 c) \times d}, E_{pos} \in \mathbb{R}^{(n+1) \times d} \quad (1)$$

As shown in [14], 1-D and 2-D positional encoding produces the same results after experiments. Therefore, to preserve the position information of patches, a simple 1-D positional encoding can be used.

3.3.2 Transformer encoder

The Transformer encoder receives the generated embedded patches z_0 with positional information. As shown in Fig. 3(b), L identical layers constitute the Transformer encoder, and each layer has two main components: (1) a multi-head self-attention block (MSA) (2), (2) A multi-layer perceptron (MLP) (3). The MLP consists of two dense layers and a GeLU activation layer between two dense layers. Each of the two components of the encoder uses residuals to skip connections, and there is a normalization layer (LN) in front of it.

$$z'_\ell = \text{MSA}(\text{LN}(z_{\ell-1})) + z_{\ell-1}, \ell = 1 \dots L \quad (2)$$

$$z_\ell = \text{MLP}(\text{LN}(z'_\ell)) + z'_\ell, \ell = 1 \dots L \quad (3)$$

$$y = \text{LN}(z_L^0) \quad (4)$$

In the last layer of the encoder, the first element in the sequence z_L^0 is gained and passed on to the head classifier, which predicts the class label.

3.3.3 Multi-head self-attention block

The function of the MSA block in the Transformer encoder is to determine the relative significance of a single patch embedding relative to other embeddings in the sequence, and the MSA block is the key part of the Transformer. This block consists of four parts: the linear layer, the self-attention layer, the concatenation layer which concatenates the outputs of the multiple attention heads, and a final linear layer, as shown in Fig. 3.

Figure 3 shows all the calculation details of the self-attention layer (SA). For each element in the input sequence, three values are generated by multiplying the element with three learned matrices U_{QKV} : Q (query), K (key), and V (value) (5). Then the dot-product between the Q vector and the K vector is calculated, and the result determines the relative importance of patches in the sequence. In the next step, the results of the dot-product are scaled and input to a *softmax*. Finally, the V vector is multiplied by the output of the

softmax to find the patch with a high attention score (6). The following equations give all operations:

$$[Q, K, V] = zU_{QKV}, U_{QKV} \in \mathbb{R}^{d \times 3D_K} \quad (5)$$

$$A = \text{softmax} \left(\frac{QK^T}{\sqrt{D_K}} \right), A \in \mathbb{R}^{n \times n} \quad (6)$$

$$SA(z) = A \cdot V \quad (7)$$

The MSA component uses the previous operations to calculate the scaled dot-product of h heads, but uses multiple Query, Key, and Value values instead of a single value. Then the results of all the attention heads are concatenated together, and finally pass through a linear layer. The following equations represent this operation:

$$MSA(z) = \text{Concat}(SA_1(z); SA_2(z); \dots SA_h(z)) W, \quad W \in \mathbb{R}^{h \cdot D_K \times D} \quad (8)$$

3.3.4 Vision Transformer variants

Different versions of the Vision Transformer are proposed in [14]: the “ViT-Base”, the “ViT-Large”, the “ViT-Huge”. The differences between these three versions are the number of layers of the encoder, the size of the hidden dimension, the size of the MLP classifier, and the number of attention heads used by the MSA layer. Table 1 shows a summary of the various versions.

Because of the success of Visual Transformers on image tasks, variants of ViT such as DeiT [46], Swin Transformer [22], CaiT [47] etc. have appeared. Touvron et al. [46] proposed a competitive convolution-free transformer called Data Efficient Image Transformer (DeiT) by training only on the ImageNet database. It uses a CNN as a teacher model and performs a knowledge distillation strategy on the Transformer to integrate the inductive bias of the Transformer. Swin Transformer [22] enhances the Transformer’s local feature extraction capability by performing local attention within a window and introducing a method of shifting window partitions for cross-window connections. CaiT [47] mainly includes the self-attention stage and the class-attention stage. In the self-attention stage, there is no class token, and in each layer, a learned diagonal matrix initialized by small values is exploited to update the channel weights dynamically, thereby offering a degree of freedom for channel adjustment. The class token is added in the class attention stage.

3.4 Data augmentation strategies

Data augmentation can be used to expand the training dataset and increase diversity. It is a simple but highly effective tool. It can be used to solve the over-fitting problem in the training process, thereby improving the robustness and generalization ability of the model.

Standard data augmentation strategies are simple geometric transformations to create new samples, such as rotation, scaling, cropping, and flipping. A new data augmentation

Table 1 Parameter statistics of each version of the Vision Transformer

Model	Numbers of Layers	Hidden Size D	MLP Layers	Heads	Numbers of Parameters
ViT-Base	12	768	3072	12	86 M
ViT-Large	24	1024	4096	16	307 M
ViT-Huge	32	1280	5120	16	632 M

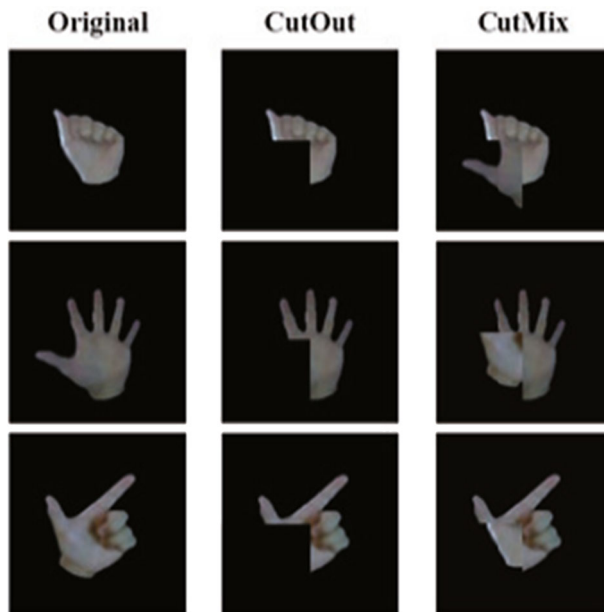


Fig. 4 Apply the CutMix and the CutOut on self-made dataset

method based on a generative model has been proposed in recent years. Among them, GANs [7] is to learn the distribution of data to generate image data similar to the original data.

The CutOut [13] and the CutMix [52] are complex technologies based on random erasing and image mixing. These technologies can generate more challenging samples for the model. In the CutOut, randomly fixed-size areas of the image are replaced by black pixels. It can encourage the model to learn from the context of the entire image, rather than relying on specific visual features. The CutMix technology works by cutting a patch of one image and replacing it with a patch of another image in the dataset. Figure 4 shows an example of the CutOut and the CutMix on the self-made gesture dataset in this article.

4 Experiments and results

4.1 Data description and data preprocessing

In the experiments of this paper, three datasets are used: (1) Dataset a: In the previous chapter, the dataset composed of images after arm removal¹; (2) Dataset b: This dataset is downloaded from the Kaggle competition platform²; (3) Dataset c: ASL Finger Spelling Dataset-A [34]. The characteristics of these three datasets are listed in Table 2, some samples of each dataset are shown in Fig. 5.

Dataset a: This dataset contains ten kinds of gestures, the custom category labels are numbers 0 to 9, 17910 images were taken by five subjects using the Kinect V2. Use the

¹<https://www.kaggle.com/datasets/zhangyu0123456/hand-dataset-first>

²<https://www.kaggle.com/grassknotted/asl-alphabet>

Table 2 Characteristics of the dataset

Dataset	Numbers of Classes	Numbers of Image per Class	Image Size
Dataset a	10	1620 ~ 1940	170 × 200
Dataset b	24	3000	200 × 200
Dataset c	24	2610 ~ 3110	— ¹

¹The sizes of these images are different

algorithm proposed in the previous chapter for these images to remove the arms and the size of each image is 512×424 . These images are then cropped to 170×200 , thus composing the Dataset a.

Dataset b: This dataset comes from the Kaggle competition platform and has 87,000 images. It contains 29 gestures including the letters A-Z of American Sign Language (ASL), del, nothing, and space. In this study, J, Z, Del, Nothing and Space are discarded. Thus, there are 24 different gestures with 3000 images in each class, and the size of each image is 200×200 .

Dataset c: ASL Finger Spelling Dataset-A contains 24 static American Sign Language (ASL) gestures (not including the letters j and z because they involve movement), and the size of each image is different. In this dataset, 65,678 images were taken by five different subjects used the Kinect V1 cameras under similar lighting and background.

Performing the image preprocessing stage before the network training can improve the overall performance [9]. The image sizes of the above three datasets are inconsistent, so first adjust the size to 224×224 uniformly. The second step is to apply the gray world color balance algorithm to perform illumination compensation. The third step is to perform morphological open operation to delete unnecessary information, that is, to perform erosion operation and then dilation operation on the image. The fourth step is to use median filter operation to filter out the salt and pepper noise in the image.

4.2 Evaluation metrics

To evaluate the performance of the proposed method, this paper used the following evaluation metrics: accuracy (Acc), precision (Pre), recall (Re) and F1-score (F1). The following is a summary of them:

- (1) **Accuracy:** Accuracy is the ratio of correctly classified samples to the total number of samples.

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} \quad (9)$$

Among them, TP, FP, TN, FN are the number of true positives, false positives, true negatives, and false negatives.

- (2) **Precision:** Precision is the ratio of the total true positives of all classes to the sum of true positives and false positives of all classes. It gives the number of correct positive predictions.

$$Precision = \frac{TP}{TP + FP} \quad (10)$$

- (3) **Recall:** Recall is the ratio of the total true positives of all classes to the total true positives and false negatives of all classes. It gives the number of correct positive

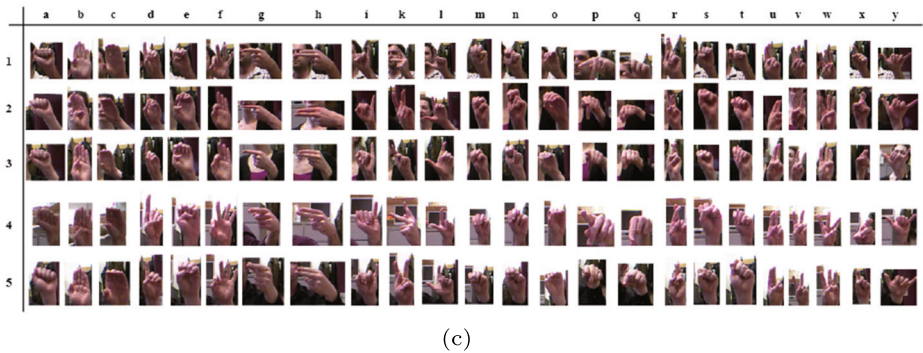
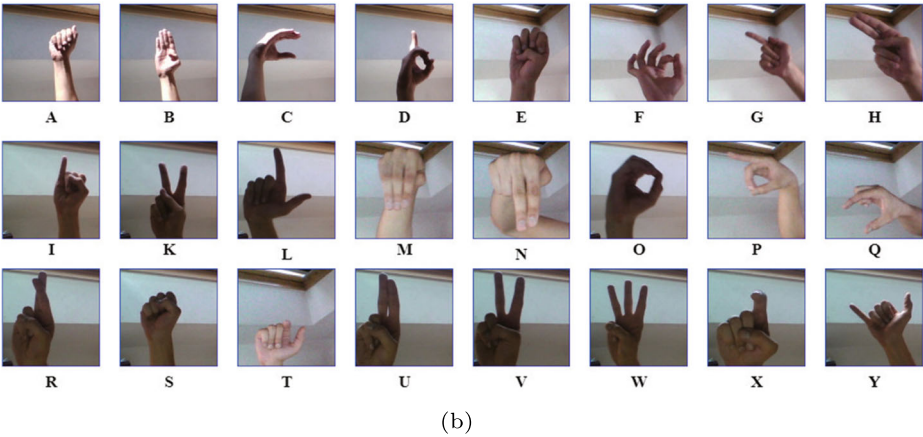
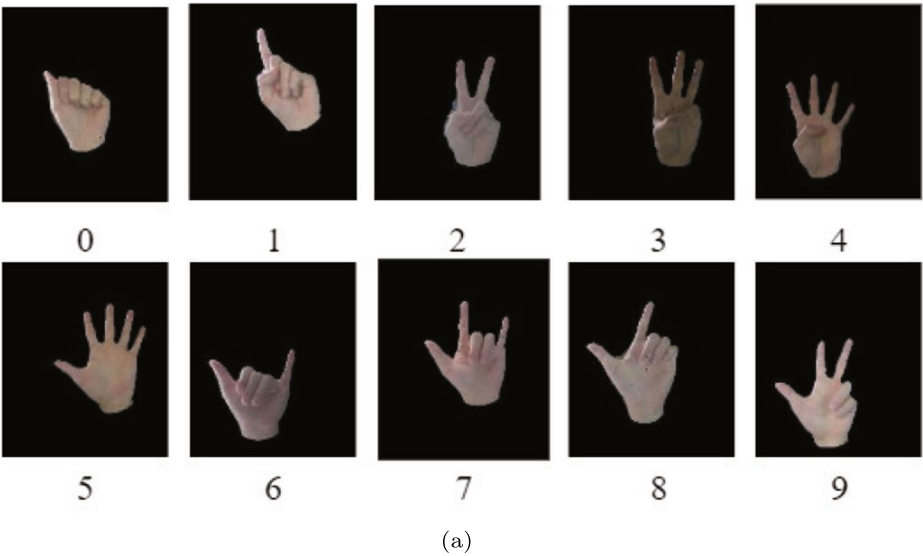


Fig. 5 Some images are from (a) Dataset a. (b) Dataset b. (c) Dataset c

predictions among all positive predictions.

$$Recall = \frac{TP}{TP + FN} \quad (11)$$

- (4) F1-score: F1 score is calculated by precision and recall. F1 score is the harmonic average of the precision and the recall.

$$F1 - score = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (12)$$

4.3 Experimental setup

In the first set of experiments, this paper used different data augmentation strategies to evaluate the Vision Transformer's performance in processing augmented data. In the second set of experiments, this paper explored the effect of changing the image size on the model's overall accuracy.

In all experiments, the ViT-Base model (ViT-B/16) was used in this paper, and the settings are as follows [14]. The model was composed of 12 encoder layers, and each layer had 12 attention heads. Its embedding dimension was 768. In this paper, a model pre-trained on Imagenet-21k was used, and then fine-tuned on Imagenet-1k. To fine-tune on the gesture datasets, this model used 50 epochs of training, and if there was no better score after waiting for 5 epochs, an early stopping strategy was implemented. The batch-size was set to 16. This paper used the Cross-Entropy loss function to learn the network's weights. The Adam method was used to optimize it, the initial learning rate was set to 1e-4 and the learning rate decay strategy of the Cosine Annealing Warm Restarts was used, and the minimum learning rate was 6e-5.

This paper used stratified k-fold cross-validation techniques [43] to conduct experiments to ensure the completeness and accuracy of the model. Stratified k-folding was one of the best ways to ensure that the data was well balanced and shuffled in each fold before splitting the data into the validation and training sets. Here, k was selected as 5 in this paper. As shown in Fig. 6, it is assumed that the dataset has five classes, and 5-fold cross-validation is used.

This paper conducted all experiments with all the following specifications on the DELL Precision 7920 Station: Central Processing Unit (CPU) Intel(R) Xeon(R) Silver 4110 CPU @2.10GHz with a RAM of 64 GB and an NVIDIA Quadro P5000 Graphical Processing Unit (GPU) with a RAM of 16 GB.

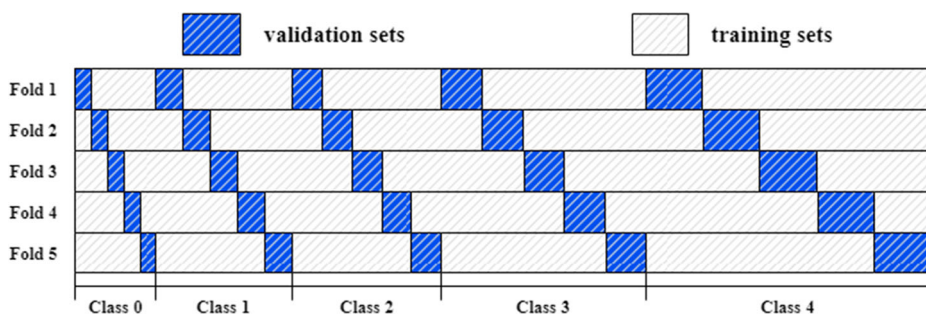


Fig. 6 Stratified five-folds cross validation. The blue parts represent the validation sets and the white parts represent the training sets

Table 3 Under the Vision Transformer pre-trained model, Acc (%), Re (%), Pre (%) and F1 (%) obtained by applying three augmentation strategies on the Dataset a

Strategy	Acc	Re	Pre	F1
Standard	99.44	99.43	99.45	99.44
CutOut	99.31	99.32	99.33	99.33
CutMix	99.40	99.39	99.40	99.39

The codes for image acquisition and foreground segmentation of the self-made dataset were implemented in the Visual Studio 2017 using the C++ code to call the library and application peripheral interface (API) provided by the Kinect V2 sensor. The arm removal process of the self-made dataset was completed on the MATLAB 2018a. The codes for the image preprocessing and network training and validation phases on all datasets were built on the Pytorch 1.9.0 framework using Python 3.6.13. The Pytorch is an open source deep neural network library written in Python. And the OpenCV interface under Python was used for image preprocessing.

4.4 Experimental results

4.4.1 The first set of experiments

This paper used the above three data augmentation strategies to evaluate the three datasets in the first set of experiments. For the standard augmentation, methods such as rotation, horizontal flipping, vertical flipping, and translation scale rotation were used. For the CutOut strategy, 8 holes were set in this paper, and the size of each hole was 10×10 . For the CutMix strategy, the parameter mixture ratio was sampled from the uniform distribution $[0, 1]$. The following three tables show the results of using the data augmentation strategies for the three datasets.

All the results are clearly shown in Tables 3, 4, 5 and Fig. 7. In summary, all data augmentation strategies have achieved close results. The Dataset b and the Dataset c use the CutMix augmentation strategy, the results obtained are better than the other two strategies, the accuracy rates are 99.37% and 96.53%, respectively. The performance of the standard augmentation strategy under the Dataset a is slightly better than the other strategies, with an accuracy rate of 99.44%. The CutOut helps the classifier to focus more on image context, such as secondary features, rather than relying heavily on primary features before making a decision. This strategy also helps with model generalization. Also, with the CutMix, the classification labels are either set to follow the majority class of the input image, or a blend ratio with the patch area. These methods can improve the robustness of the model to input

Table 4 Under the Vision Transformer pre-trained model, Acc (%), Re (%), Pre (%) and F1 (%) obtained by applying three augmentation strategies on the Dataset b

Strategy	Acc	Re	Pre	F1
Standard	98.45	98.45	98.48	98.47
CutOut	98.41	98.41	98.45	98.43
CutMix	99.37	99.37	99.38	99.38

Table 5 Under the Vision Transformer pre-trained model, Acc (%), Re (%), Pre (%) and F1 (%) obtained by applying three augmentation strategies on the Dataset c

Strategy	Acc	Re	Pre	F1
Standard	95.69	95.68	95.82	95.75
CutOut	96.29	96.27	96.39	96.33
CutMix	96.53	96.51	96.67	96.59

corruption. Simultaneously, the CutMix can stabilize the training process and improve the visual Transformer [46]. So the CutMix and the CutOut can get slightly better results.

4.4.2 The second set of experiments

In the second set of experiments, this paper further explored the impact of the input image size on the model's performance. Therefore, this paper used two different image sizes of 224×224 and 384×384 to conduct the same experiments. The results of the experiments are shown in Table 6 below. The results show that when the image size is increased, the model's accuracy is improved, and the Dataset c is even improved by 1.33%. Regardless of the image resolution, the model's patch size is still 16, so if the image resolution is larger, the number of patches will increase. Further, more detailed context information will be obtained during the training process. However, as the image size increases, the training time of the model also increases. In the experiments of this paper, the training time of the image size of 384×384 is three times longer than the image size of 224×224 .

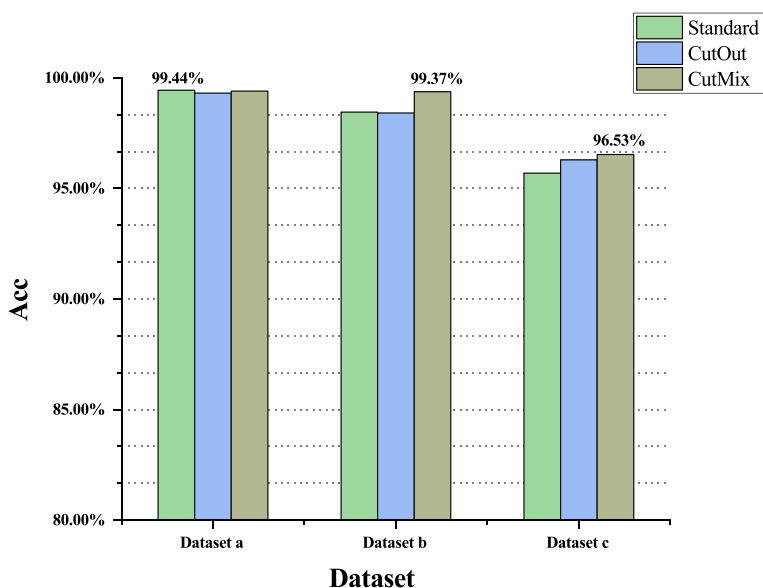
**Fig. 7** All the results

Table 6 The accuracy (%) obtained by using different image sizes on different datasets

Dataset	224 × 224	384 × 384
Dataset a	99.44	99.64
Dataset b	99.37	99.38
Dataset c	96.53	97.86

4.4.3 Comparison with existing methods

In this subsection, the method proposed in this paper is compared with several current advanced methods. These methods are based on convolutional neural network implementation methods, including Fine-tuned VGG16, Fine-tuned VGG19, Fine-tuned MobileNet and several Baseline CNNs. There are also some visual Transformer based methods which are the DeiT-B, the Swin-B and the CaiT-S24. The method proposed in this paper use the ViT model, and the size of the patches is 16×16 , pre-trained on Imagenet-21k and fine-tuned with images of size 224×224 and 384×384 . All results are shown in Table 7 and Fig. 8. As shown in the table, the method in this paper is higher than other methods. Because skip connections are also included in the ViT model, skip connections have a greater impact on the model than CNNs (ResNet), and greatly affect the representation and similarity of features. Through pre-trained with a large amount of data, the ViT can learn high-quality intermediate features [14].

4.4.4 Results

In the previous experiments, this paper obtained three results: (1) In Tables 3, 4 and 5, the complex data augmentation strategies improved the accuracy of the algorithm to a certain extent in the process of training the ViT model. (2) Larger resolution can also improve

Table 7 Comparison with other methods

Method	FLOPs	Params	Acc		
			Dataset a	Dataset b	Dataset c
Baseline CNN1 [5]	0.2G	13M	94.84	95.14	92.88
Baseline CNN2 [18]	0.6G	48M	97.75	94.71	95.79
ADCNN [1]	0.3G	12M	96.02	97.79	96.01
CNN-TPE [10]	0.1G	9M	98.25	95.74	94.70
Fine-tuned MobileNet [10]	0.6G	3M	98.13	95.38	93.67
Fine-tuned VGG16 [10]	15.5G	134M	97.69	94.66	91.20
Fine-tuned VGG19 [19]	19.6G	140M	98.35	97.45	94.67
DeiT-B(224 ²) [46]	17.7G	86M	99.40	99.28	96.34
Swin-B(224 ²) [22]	15.5G	87M	99.20	99.18	96.48
CaiT-S24(224 ²) [47]	9.4G	47M	99.3	99.16	95.53
Proposed Method(224 ²)	17.6G	86M	99.44	99.37	96.53
Proposed Method(384 ²)	55.5G	86M	99.64	99.38	97.86

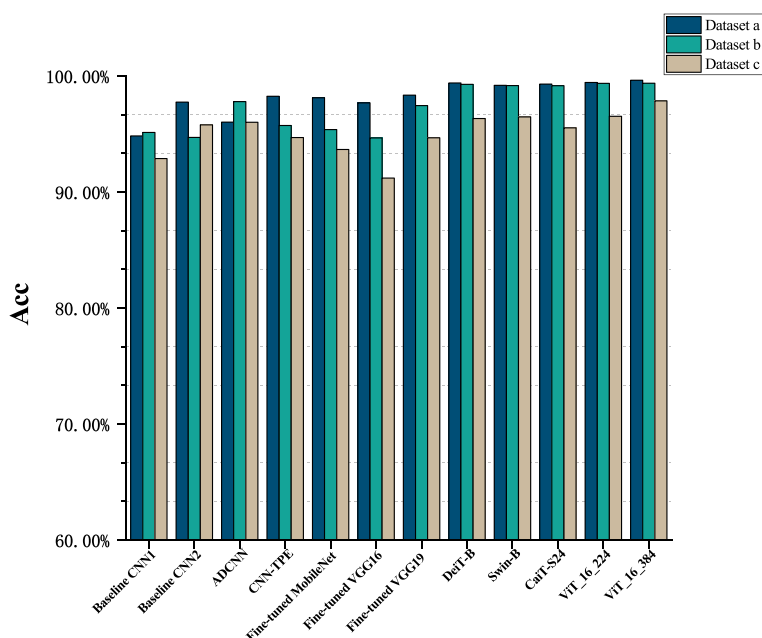


Fig. 8 Comparison with other methods

the accuracy of the algorithm. This is due to the fact that during the ViT model training, the larger the resolution, the larger the number of patches. (3) In Table 7, it can be seen that the visual Transformers have higher accuracy than the CNN model, because the visual Transformers can show better robustness in the case of complex data augmentation strategies.

5 Conclusion

This paper uses the depth threshold of the Kinect V2 camera to segment the foreground part of interest. The depth information of the obtained image is mapped to the color space to obtain the gesture image containing the arm. Next, the eight-connected discrimination algorithm and the distance transformation algorithm remove the redundant arm part. Finally, a gesture image dataset that does not contain the arm is obtained.

This paper proposes a static hand gesture recognition method based on the Vision Transformer. This model uses a self-attention mechanism to capture the long-term dependencies between image patches. The evaluation was conducted on the self-made dataset and the other two datasets, and certain data augmentation strategies were used to improve the classification accuracy of gesture images. The recognition accuracy obtained was 99.44%, 99.37% and 96.53%, respectively. At the same time, a larger image resolution was used for evaluation, and the recognition accuracy obtained was 99.64%, 99.38% and 97.86%, respectively. The accuracy is improved, but the difficulty of training increases and the training time becomes longer. Compared with the method based on CNN, the recognition accuracy of this model is higher.

Although this paper has demonstrated the usefulness of the ViT for hand gesture recognition tasks, there are some limitations. First of all, there are few literatures on the use of visual Transformers for hand gesture recognition tasks, so this paper only refers to the application of the CNN models in hand gesture recognition tasks. Second, the ViT model has more parameters than the most CNN models, which makes training more difficult and time-consuming, and a small model is more beneficial for deploying gesture recognition. Therefore, the future research direction is mainly to apply more visual Transformers to hand gesture recognition tasks in order to obtain better generalization models. And reduce the parameters of the ViT model, making it more favorable for deployment.

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Data Availability The datasets used to support the findings of this study could be found freely here:

Dataset a: <https://www.kaggle.com/datasets/zhangyu0123456/hand-dataset-first>

Dataset b: <https://www.kaggle.com/grassknotted/asl-alphabet>

Dataset c: <https://empslocal.ex.ac.uk/people/staff/np331/index.php?section=FingerSpellingDataset>

Declarations

Conflict of Interests The authors declare that they have no conflict of interest.

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